



University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

Programming Fundamental ~ Lab Project

Project Information

Project Name	Quiz Generator System
Course Title	Programming Fundamental - PF LAB
Submission Date	11-January -2026
Faculty	Faculty Of Information Technology



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Program Description

The **Quiz Generator System** is a **menu-driven C++ console application** that allows users to take quizzes on multiple subjects at different difficulty levels.

Key characteristics of the program:

- Supports multiple subjects: **Programming Fundamentals (PF)**, **English (Eng)**, **Mathematics (Math)**
- Reads questions from external **text files**
- Randomizes questions for each quiz attempt
- Calculates scores in real-time and applies **negative marking** for advanced difficulty
- Saves quiz results in `results.txt`
- Provides colorful console interface and sound feedback for enhanced user experience

The system demonstrates practical **Programming Fundamentals concepts**, including file handling, structures, arrays, loops, input validation, and modular programming.

Key Feature:

1. Multi-Subject Quiz System

- Supports PF, Eng, and Math
- Each subject has **separate question files**
- Questions are **organized per subject and difficulty level**

2. Difficulty Levels

- Beginner
- Intermediate
- Advanced
- Advanced difficulty includes **negative marking** (−10 points for wrong answers)

3. File-Based Question Management

- Questions stored in external `.txt` files
- File paths are determined dynamically based on subject and difficulty
- Easy to **update or add questions** without modifying the program

4. Structured Question Handling

- Uses a `struct Question` to store:
 - Question statement
 - Four multiple-choice options
 - Correct answer
- Questions displayed clearly with A/B/C/D options
- **Randomized question order** for each quiz attempt



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5. Score Calculation System

- Correct answer → **+10 points**
- Wrong answer in Advanced → **-10 points**
- Displays **current score** after each question
- Shows **final score** at the end

6. Result Storage and Viewing

- Saves results in `results.txt`
- Records include: Player Name | Subject | Difficulty | Score
- Users can **view previous results** from the menu

7. User-Friendly Menu Interface

- Options:
 - Start Quiz
 - Instructions
 - View Results
 - Exit
- Uses **`system("cls")`** for clean transitions
- Prompts and colors make the interface easy to follow

8. Input Validation

- Validates **player name** (letters & spaces only)
- Validates **menu choice** and **difficulty/subject selection**
- Accepts **only valid answer letters (A–D)**

9. Enhanced User Interaction

- Uses **`SetConsoleTextAttribute`** to show colored text
- Uses **`Beep()`** to provide sound feedback for correct/incorrect answers
- Sleep delays (`Sleep()`) enhance readability between questions

Features Used:

1. Core C++ Concepts

- **Structures (`struct`)** – store questions, options, and answers
- **Arrays** – handle multiple questions
- **Loops (`for`, `while`, `do-while`)** – quiz navigation & question display
- **Conditional statements (`if`, `else`)** – input validation & scoring
- **Functions** – modularize code (`startQuiz()`, `instructions()`, `viewResults()`, etc.)

2. File Handling

- `ifstream` – reads questions from `.txt` files
- `ofstream` – saves quiz results to `results.txt`
- File-based system ensures **data persistence**



3. Standard Libraries

- `<iostream>` – input/output
- `<fstream>` – file handling
- `<string>` – string manipulation
- `<cctype>` – character handling
- `<cstdlib>` & `<ctime>` – randomization of questions

4. System Integration

- `system("cls")` – clear console for clean transitions
- `Sleep()` – delay between messages/questions
- `Beep()` – sound feedback for correct/wrong answers
- `SetConsoleTextAttribute()` – colored output for better UX

Core Mechanics:

• Main Menu

- User chooses: Start Quiz, Instructions, View Results, Exit

• Player Info

- User enters **name** (validated for letters and spaces)

• Subject Selection

- Choose one: PF, Eng, Math

• Difficulty Selection

- Choose one: Beginner, Intermediate, Advanced

• Question Loading

- Program selects **file path** based on subject & difficulty
- Loads questions into `struct` array
- Questions are **shuffled randomly**

• Quiz Execution

- Displays **10 random questions** (or fewer if less available)
- Shows options (A–D)
- Takes user input and validates it
- Updates score (+10 / –10) and shows current score
- Uses color and sound feedback

• Result Display & Storage

- Shows final score and quiz summary



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- Saves result to `results.txt`
- Users can later view **all previous results**